en(B)

Twenty-Third International Linguistics Olympiad

Bucharest (Romania), 26 July – 2 August 2026

Individual Contest Problems

Rules for writing out the solutions

Do not copy the problems. Write down your solution to each problem on a separate sheet or sheets. On each sheet indicate the number of the problem, the number of your seat and your surname. If you do not do this, your work may be mislaid or misattributed.

Unless stated differently, you should describe any patterns or rules that you identified in the data. Otherwise your solution will not be awarded full marks.

Problem 1 (20 points). Here are some words in Central Yup'ik along with their phonetic transcriptions and translations. All of them are pronounced in the Norton Sound (NS) dialect:

qasgulek	[ˈqaz.ɣu.ɮəkʰ]	*ribbon seal*
nanevpak	[na.ˈnəf.pakʰ]	*big lake*
tep'liuq	[ˈtəp.ɮi.uqʰ]	*he is making aged fish-heads*
nav'pagtuq	[ˈnav.pax.tuqʰ]	*it really broke*
qamiq'vak	[qa.ˈmiq.vakʰ]	*a type of moth*
nat'rarkaq	[ˈnat.ʁaχ.kaqʰ]	*sole material for skin boots*
calugun	[ca.ˈɮuː.ɣun]	*skin scraper*
kaal'taaq	[ka.ˈaɮ.ta.aqʰ]	*playing card*
ullelluku	[u.ˈɬəɬ.ɬu.ku]	*turning it inside out*
ilquineq	[ˈiɬ.qu.ˈiː.nəqʰ]	*skull*
kaltuuvvilaq	[ˈkaɬ.tu.ˈuː.fi.ɮaqʰ]	*potato*
kayangurngalnguq	[ka.ˈjaː.ŋuʁ.ˈŋaɮ.ŋuqʰ]	*oval thing*
assikekeka	[a.ˈsiː.kə.ˈkək.ka]	*I like him*
paluteqerru	[pa.ˈɮuː.tə.ˈqəχ.χu]	*please close it*
iggiggiayuli	[i.ˈxiː.xi.ˈaː.ju.ɮi]	*great horned owl*
ilalkumalria	[i.ˈɮaɬ.ku.ˈmaɮ.ʁi.a]	*victim*
apepailgaku	[a.ˈpəp.pa.ˈiɮ.ɣa.ku]	*before asking him*
yuarutlertangqertuq	[ju.ˈaː.ʁutʰ.ˈɬəχ.taŋ.ˈqəχ.tuqʰ]	*there is an old song*
tutgarrlugaq	[ˈtutʰ.xaχ.ˈɬuː.ɣaqʰ]	*grandchild*
cuukvagtalek	[cu.ˈukʰ.fax.ˈtaː.ɮəkʰ]	*place with many pike (fish)*
cangatenrituten	[ca.ˈŋaː.tən.ˈʁiː.tu.tən]	*there is nothing wrong with you*
cungagcessvakartat	[cu.ˈŋax.cəs.ˈfaː.kaχ.tatʰ]	*how green they are!*

Out of the forms above, only the last four are pronounced differently in General Central Yup'ik (GCY), another dialect of the language. Here are the phonetic transcriptions of these forms, as pronounced in the GCY dialect:

tutgarrlugaq	[ˈtutʰ.ˈxaχ.ɬu.ɣaqʰ]	*grandchild*
cuukvagtalek	[cu.ˈukʰ.ˈfax.ta.ɮəkʰ]	*place with many pike (fish)*
cangatenrituten	[ca.ˈŋaː.ˈtən.ʁi.ˈtuː.tən]	*there is nothing wrong with you*
cungagcessvakartat	[cu.ˈŋax.ˈcəs.fa.ˈkaχ.tatʰ]	*how green they are!*

(a) Write in Central Yup'ik:

1.	[ˈməɬ.qu.ˈɮəɮ.ɮi.ˈuχ.ta]	*wildlife officer*
2.	[qa.ˈjaː.pix.ˈkaː.ni]	*his own future authentic kayak*
3.	[ta.ˈɬiː.χaqʰ]	*hook used in kayak*
4.	[ˈkəx.sa.ˈiɬ.ku.taqʰ]	*muzzle for dog*
5.	[ˈcup.ɣa.ˈɬax.tuqʰ]	*he suddenly felt homesick*

For forms that are valid in only one of the two dialects (NS or GCY), indicate which dialect is represented.

(b) Write the phonetic transcriptions of the forms below:

6.	**aartartuq**	*he tends to be wary*
7.	**akekataki**	*ouch!*
8.	**negavgun**	*root used as scrubber in bath*
9.	**atasuak**	*summer trousers*
10.	**ner'ciquq**	*he will eat*
11.	**alikenritaqa**	*I do not fear it*
12.	**neqlinrrarmek**	*from fish cut up just now*

For transcriptions that differ across dialects, write both transcriptions and indicate the dialect for each of them.

⚠ The Central Yup'ik language belongs to the Eskimo-Aleut family. It is spoken by approx. 10,000 people in southwestern Alaska in the United States of America.

The mark [ː] indicates that the vowel is long. The mark [.] separates syllables. The mark [ˈ] goes before the stressed syllable(s).

[ʰ] indicates aspiration of the preceding consonant.

[ə] = *a* in *about*.

[c] ≈ *tch* in *match*.

[j] = *y* in *you*.

[ɬ] = *s* in *sing*, but with the tongue positioned as for *l*; [ɮ] = [ɬ], but with vocal cord vibration.

[ŋ] = *ng* in *hang*.

[q] = *c* in *car*, but further back in the mouth.

[x] = *ch* in *Bach*; [ɣ] = [x], but with vocal cord vibration.

[χ] = [x], but further back in the mouth; [ʁ] = [χ], but with vocal cord vibration.

—Lai Otsuka (consultant: Sally Samson)